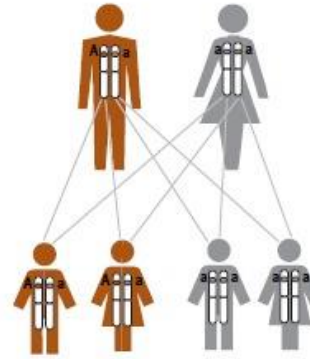


# Inheritance of DNA variation

Why important?

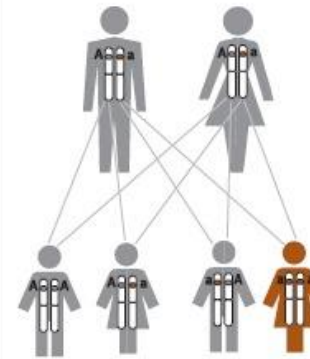
1. Genetic disease of humans
2. Breeding of desirable characters in cultivated plants and animals

■ Affected ■ Unaffected



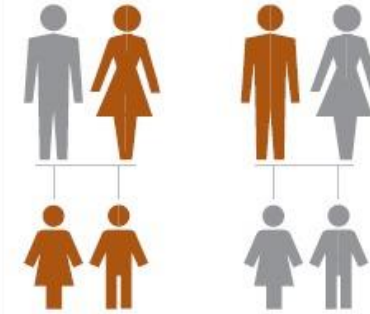
**AUTOSOMAL DOMINANT**

- Individuals carrying one mutated copy of a gene in each cell will be affected by the disease
- Each affected person usually has one affected parent
- Tends to occur in every generation of an affected family



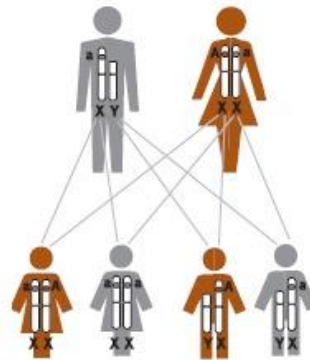
**AUTOSOMAL RECESSIVE**

- Affected individuals must carry two mutated copies of a gene
- Parents of affected individual are usually unaffected and each carry a single copy of the mutated gene (known as carriers)
- Not typically seen in every generation.



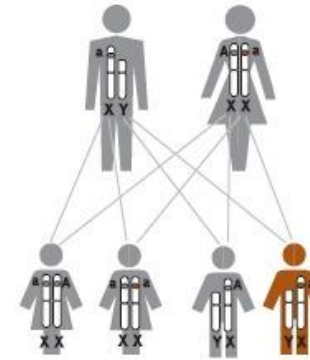
**MITOCHONDRIAL**

- Only females can pass on mitochondrial conditions to their children (maternal inheritance)
- Both males and females can be affected
- Can appear in every generation of a family



**X-LINKED DOMINANT**

- Females are more frequently affected than males
- Fathers cannot pass X-linked traits to their sons (no male-to-male transmission)



**X-LINKED RECESSIVE**

- Males are more frequently affected than females
- Families with an X-linked recessive disorder often have affected males, but rarely affected females, in each generation
- Both parents of an affected daughter must be carriers
- Only mother must be carrier of affected son (fathers cannot pass X-linked traits to theirs)

# Mendelian Genetics

Gregor Johann Mendel (1822-1884),  
Austrian investigator,  
who carried out  
the first **scientific experiments**  
leading to the  
**discovery of patterns of inheritance**  
from parents to offspring



# What did Mendel do?

these unnatural characters are due to mutations in some genes

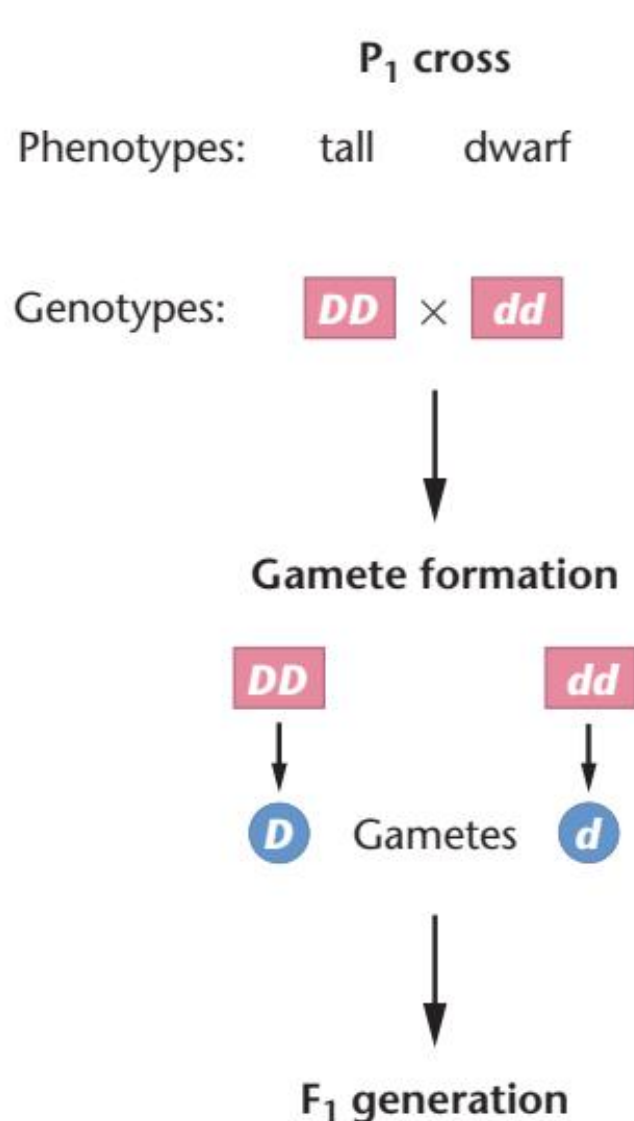
Results of crosses between contrasting traits/characters

| Character       | Contrasting traits |                                                                                                                                                                           | F <sub>1</sub> results | F <sub>2</sub> results      | F <sub>2</sub> ratio |
|-----------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------|----------------------|
| Seed shape      | round/wrinkled     |  X  =  | all round              | 5474 round<br>1850 wrinkled | 2.96:1               |
| Seed color      | yellow/green       |        | all yellow             | 6022 yellow<br>2001 green   | 3.01:1               |
| Pod shape       | full/constricted   |        | all full               | 882 full<br>299 constricted | 2.95:1               |
| Pod color       | green/yellow       |        | all green              | 428 green<br>152 yellow     | 2.82:1               |
| Flower color    | violet/white       |        | all violet             | 705 violet<br>224 white     | 3.15:1               |
| Flower position | axial/terminal     |      | all axial              | 651 axial<br>207 terminal   | 3.14:1               |
| Stem height     | tall/dwarf         |   | all tall               | 787 tall<br>277 dwarf       | 2.84:1               |

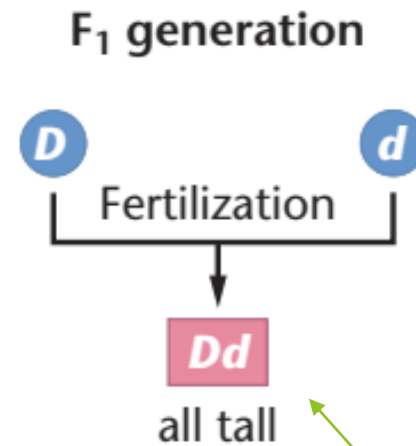
Only one of the contrasting traits was obtained in the first generation of crosses (F<sub>1</sub>)

The traits segregated in approximately 3:1 ratio in the second generation of the crosses (F<sub>2</sub>)

# Monohybrid cross: explanation of the results



Looks easy



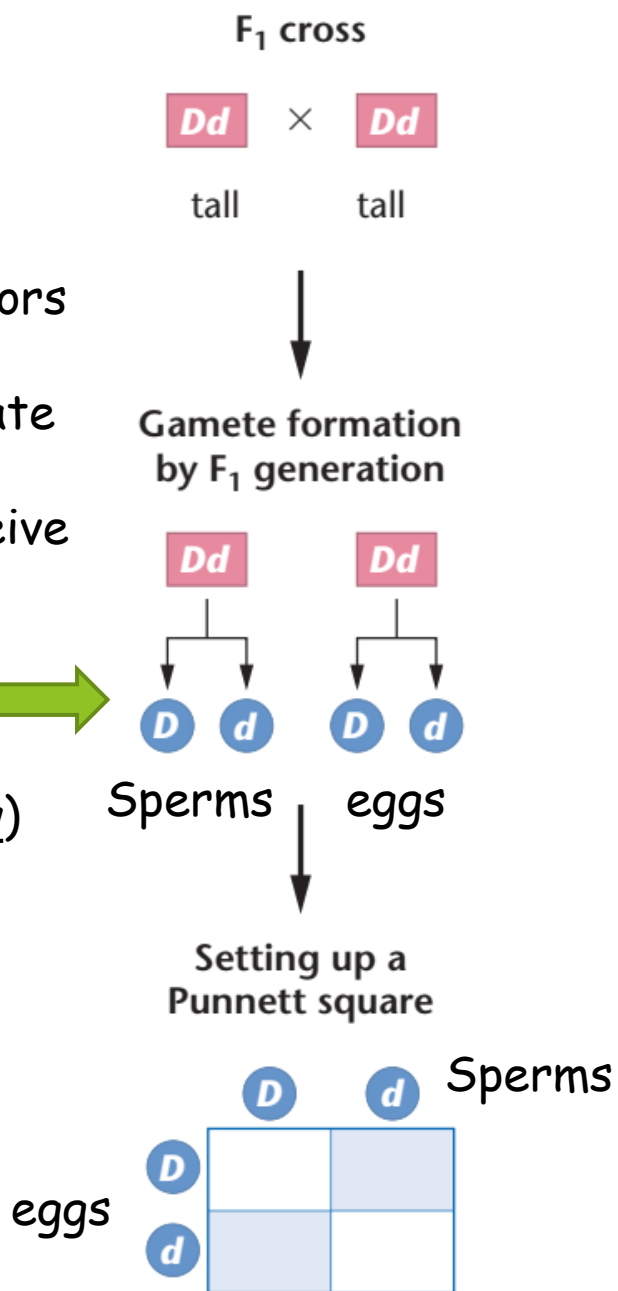
The results of Mendel's crosses can only be explained if we assume:

1. There are two **unit factors** (variants/alleles of a gene) for each trait and they are in **pairs** e.g. **DD** and **dd** for plant height (tall/dwarf)
2. One unit factor(allele) is dominant over the other e.g. **D** is dominant over **d** (so it masks the effect of **d**; hence all individuals of **F<sub>1</sub>** are tall)

The above two are the **first 2 laws of Mendel**

Pairs of unit factors (alleles) separate/segregate randomly so that each gamete receive one or the other with equal probability

(Mendel's 3<sup>rd</sup> law)



# Mendel's laws explained by modern knowledge of cytogenetics

Cytogenetics is the study of the behavior of chromosomes and the genes they contain at different stages of cell division

1<sup>st</sup> & 2<sup>nd</sup> Laws. **Unit factors** in pairs (first meiotic prophase)

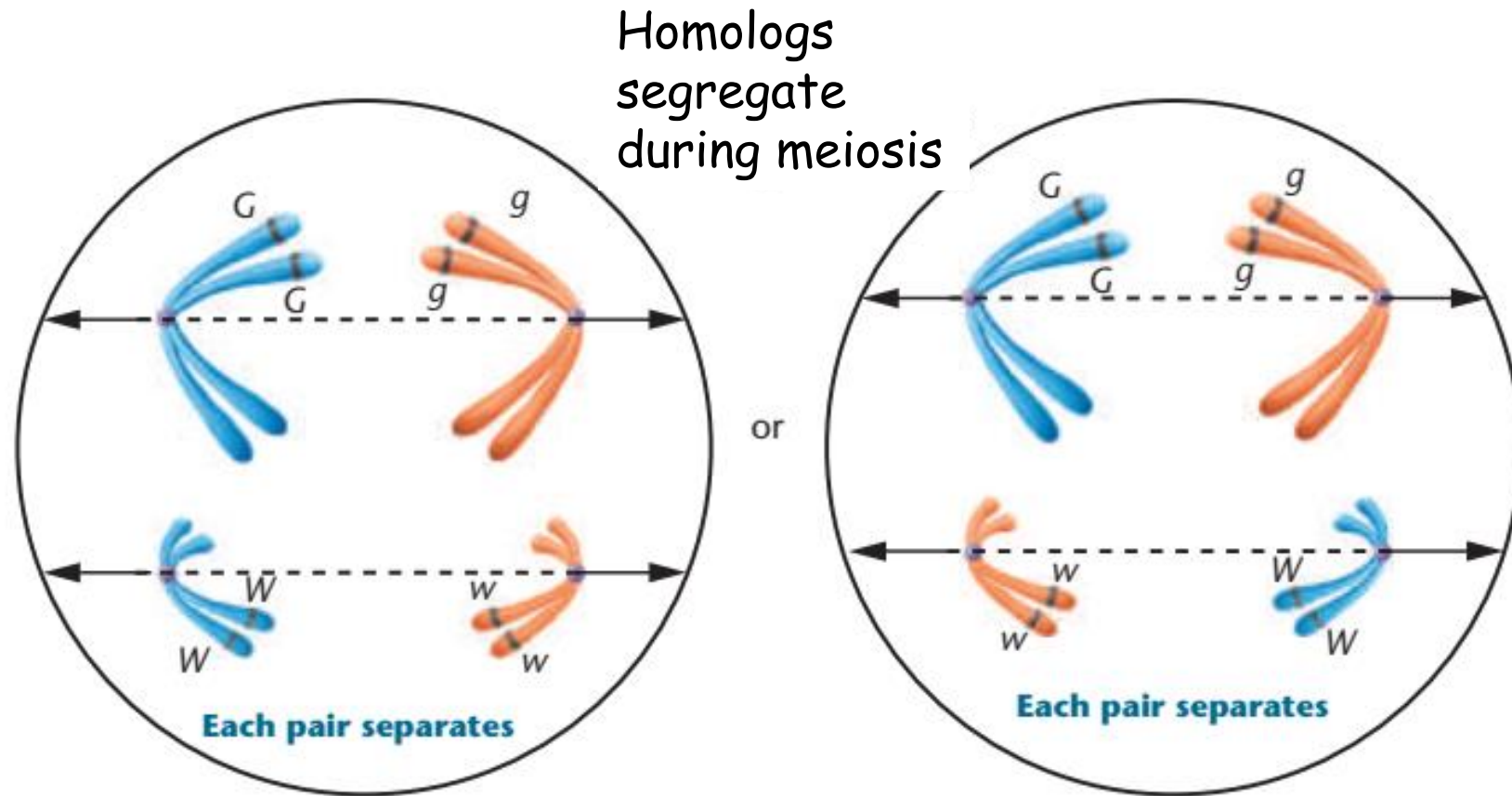


**Genes are part of chromosomes**

Each gene has 2 alleles (at least in this case)

Gene product of one allele overrides the other

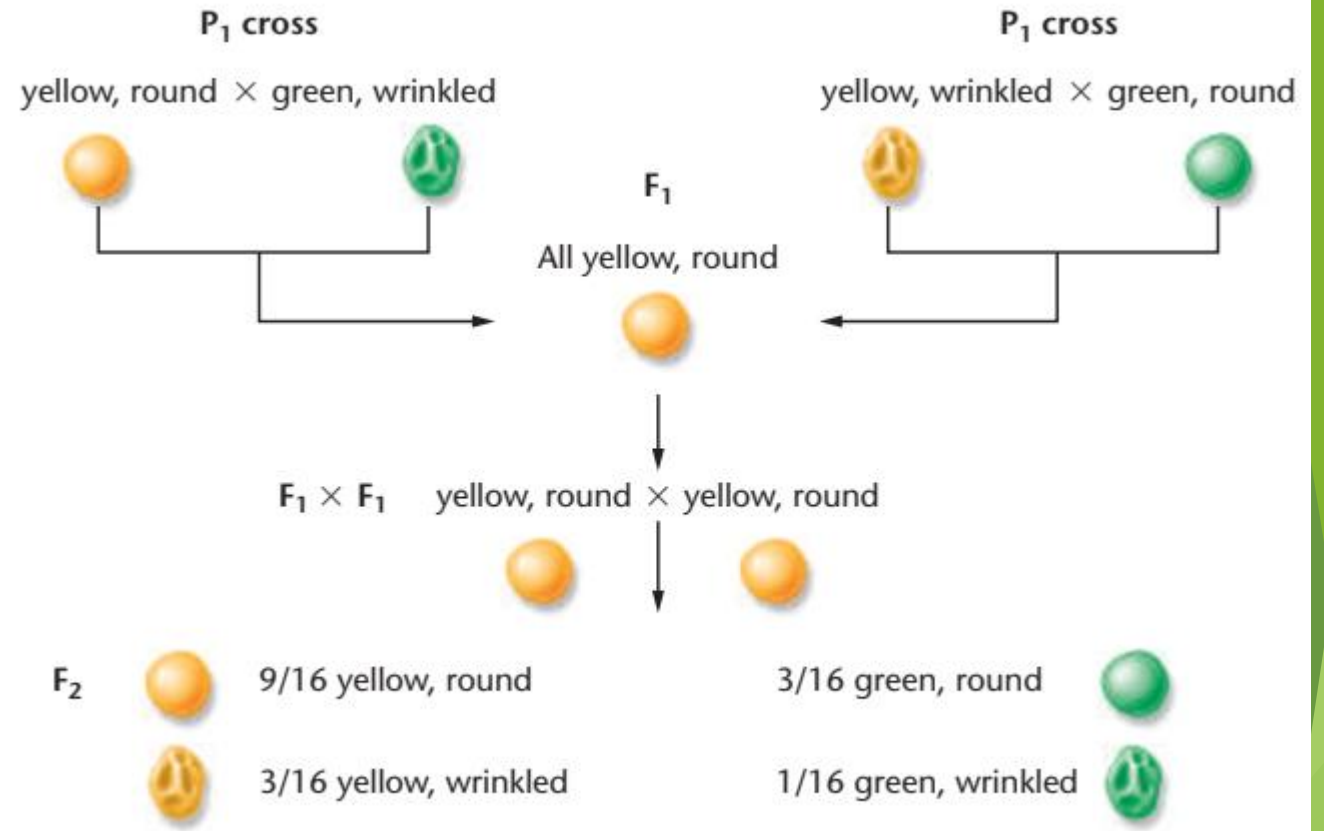
### 3<sup>rd</sup> Law. Segregation of unit factors during gamete formation (first meiotic anaphase)



# Dihybrid cross

When 2 pairs of characters are involved, the segregating pairs of unit factors assort independently of each other

(Mendel's 4<sup>th</sup> law)



# Dihybrid cross

Gametes with all possible combinations of genotypes are produced

i.e., in a gamete, the probability of getting  $G/g$  is not influenced by the probability of getting  $W/w$  (**Independent assortment**)

F<sub>2</sub> Genotypic ratio

1/16 GGWW

2/16 GGWw

2/16 GgWW

4/16 GgWw

F<sub>2</sub> Phenotypic ratio

9/16 yellow, round

1/16 GGww

2/16 Ggww

3/16 yellow, wrinkled

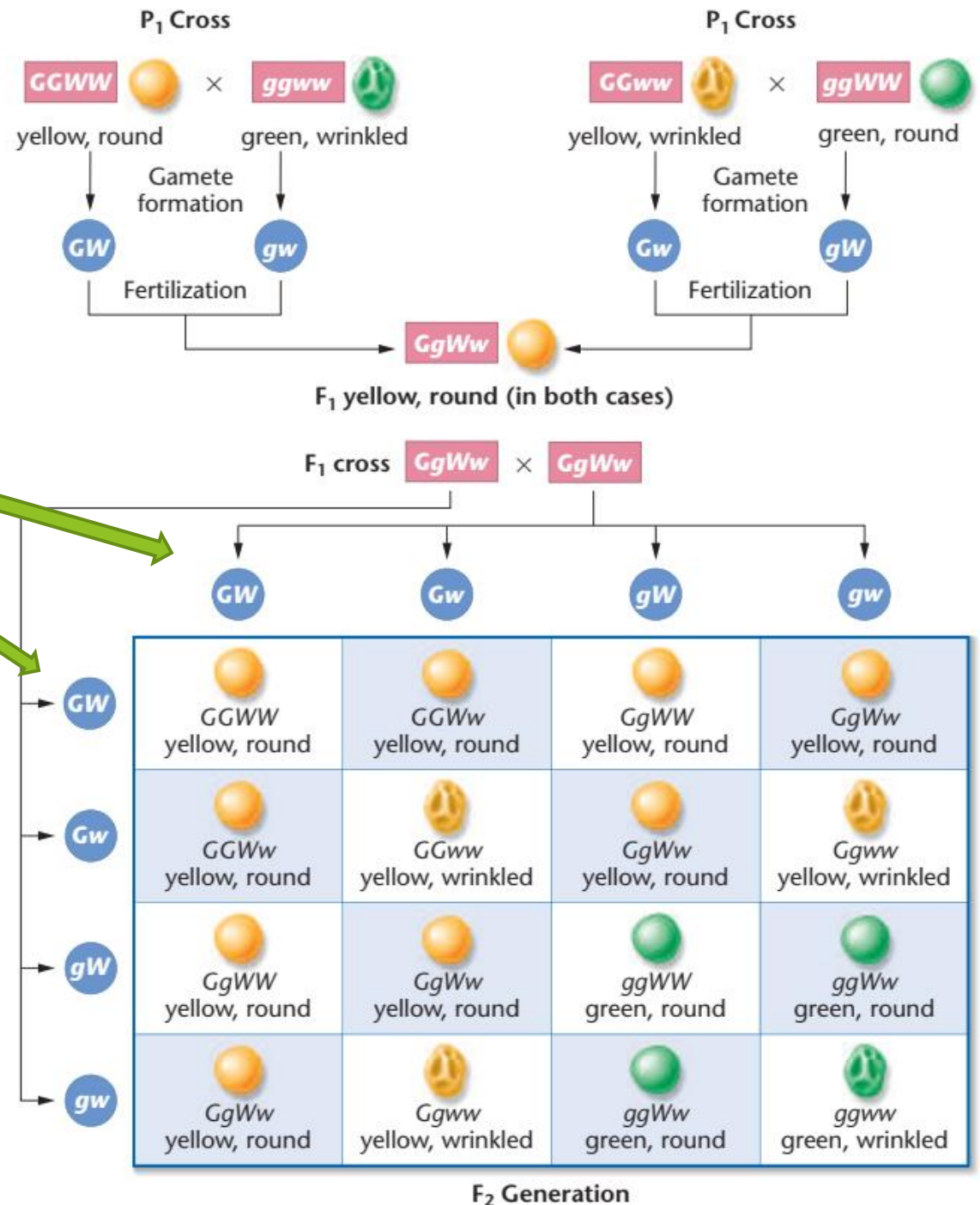
1/16 ggWW

2/16 ggWw

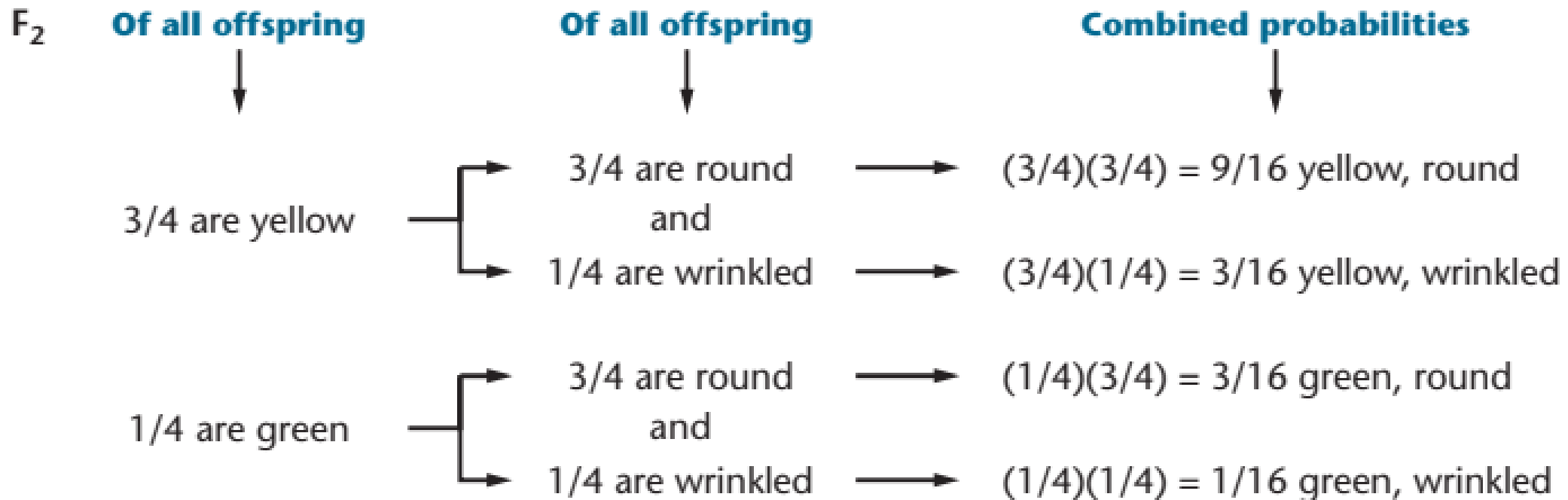
3/16 green, round

1/16 ggww

1/16 green, wrinkled



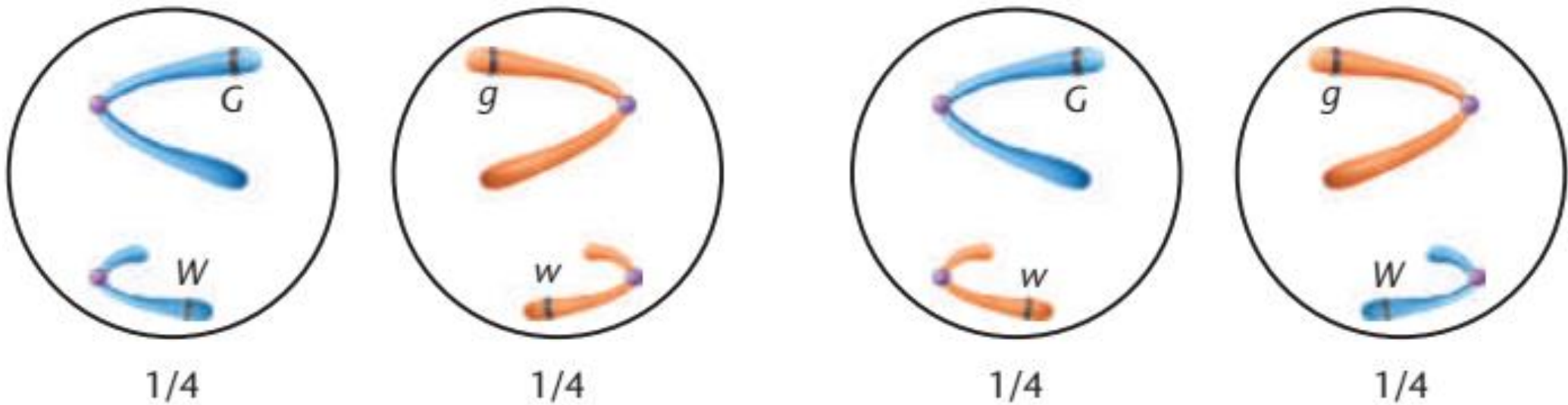
# Outcome of the dihybrid cross can be explained by the product law of probability



product of independent outcomes  
(probability of being yellow or green)  $\times$  (probability of being round or wrinkled)

4<sup>th</sup> Law. Independent assortment of paired unit factors  
(finally after many meiotic events in telophase-II)

Nonhomologous chromosomes assort independently



All possible gamete combinations formed with equal probability

# Molecular explanation of Mendel's traits (actual genes)



*Starch branching enzyme (SBEI) gene*



round



*Starch branching enzyme (SBEI) gene  
dysfunctional due to a transposon insertion*



wrinkled

transposons are jumping genes



# Chi square analysis and influence of chance on genetic experiment data

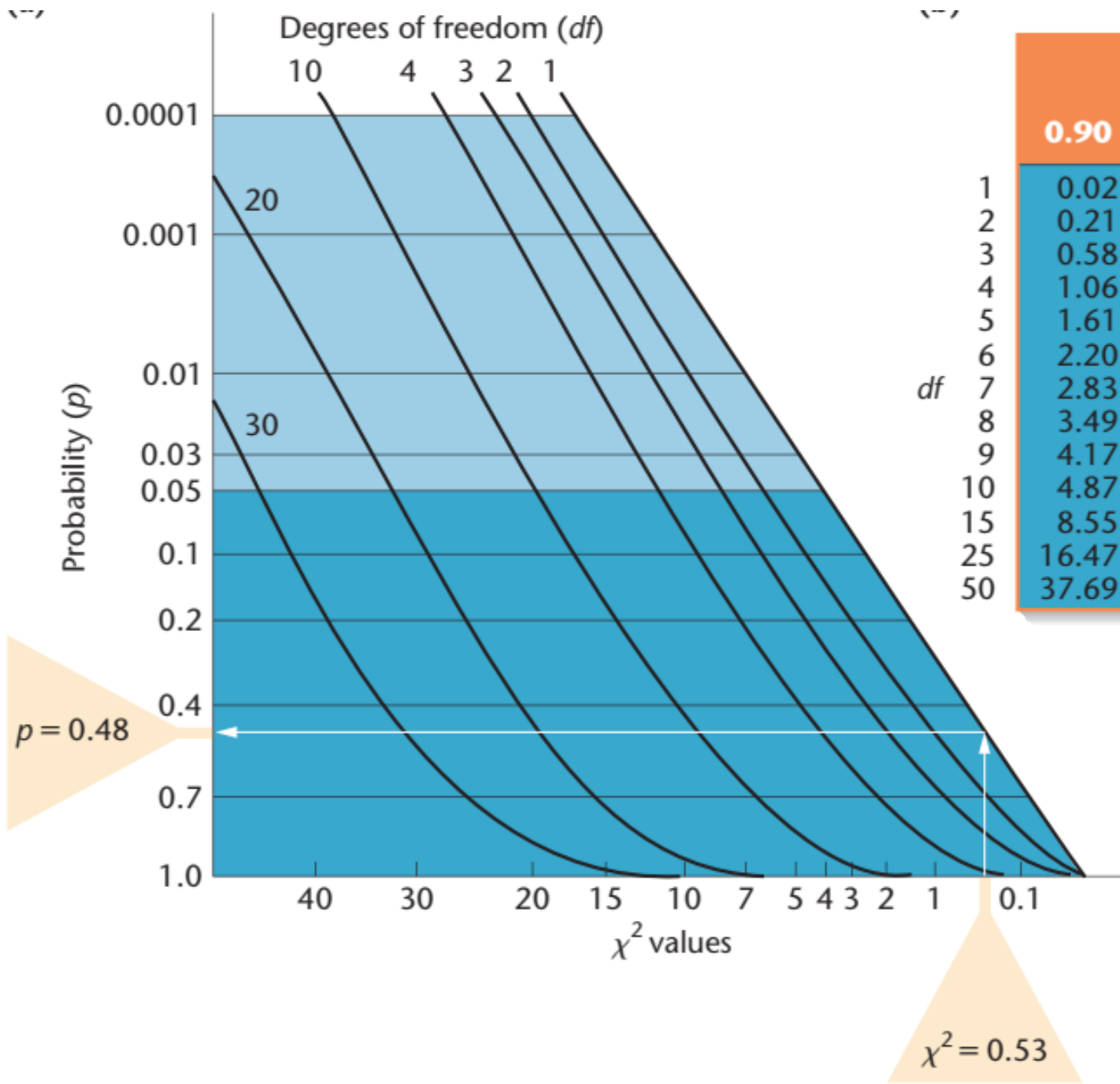
$$\chi^2 = \sum \frac{(o - e)^2}{e}$$

## (a) Monohybrid

| Cross Expected Ratio | Observed (o) | Expected (e)    | Deviation (o - e) | Deviation (d <sup>2</sup> ) | d <sup>2</sup> /e     |
|----------------------|--------------|-----------------|-------------------|-----------------------------|-----------------------|
| 3/4                  | 740          | 3/4(1000) = 750 | 740-750 = -10     | (-10) <sup>2</sup> = 100    | 100/750 = 0.13        |
| 1/4                  | <u>260</u>   | 1/4(1000) = 250 | 260-250 = +10     | (+10) <sup>2</sup> = 100    | 100/250 = <u>0.40</u> |
|                      | Total = 1000 |                 |                   |                             | $\chi^2 = 0.53$       |
|                      |              |                 |                   |                             | $p = 0.48$            |

## (b) Dihybrid

| Cross Expected Ratio | Observed (o) | Expected (e) | Deviation (o - e) | Deviation (d <sup>2</sup> ) | d <sup>2</sup> /e |
|----------------------|--------------|--------------|-------------------|-----------------------------|-------------------|
| 9/16                 | 587          | 567          | +20               | 400                         | 0.71              |
| 3/16                 | 197          | 189          | +8                | 64                          | 0.34              |
| 3/16                 | 168          | 189          | -21               | 441                         | 2.33              |
| 1/16                 | <u>56</u>    | 63           | -7                | 49                          | <u>0.78</u>       |
|                      | Total = 1008 |              |                   |                             | $\chi^2 = 4.16$   |
|                      |              |              |                   |                             | $p = 0.26$        |



|    | Probability ( $p$ ) |       |       |       |       |       |
|----|---------------------|-------|-------|-------|-------|-------|
|    | 0.90                | 0.50  | 0.20  | 0.05  | 0.01  | 0.001 |
| 1  | 0.02                | 0.46  | 1.64  | 3.84  | 6.64  | 10.83 |
| 2  | 0.21                | 1.39  | 3.22  | 5.99  | 9.21  | 13.82 |
| 3  | 0.58                | 2.37  | 4.64  | 7.82  | 11.35 | 16.27 |
| 4  | 1.06                | 3.36  | 5.99  | 9.49  | 13.28 | 18.47 |
| 5  | 1.61                | 4.35  | 7.29  | 11.07 | 15.09 | 20.52 |
| 6  | 2.20                | 5.35  | 8.56  | 12.59 | 16.81 | 22.46 |
| 7  | 2.83                | 6.35  | 9.80  | 14.07 | 18.48 | 24.32 |
| 8  | 3.49                | 7.34  | 11.03 | 15.51 | 20.09 | 26.13 |
| 9  | 4.17                | 8.34  | 12.24 | 16.92 | 21.67 | 27.88 |
| 10 | 4.87                | 9.34  | 13.44 | 18.31 | 23.21 | 29.59 |
| 15 | 8.55                | 14.34 | 19.31 | 25.00 | 30.58 | 37.30 |
| 25 | 16.47               | 24.34 | 30.68 | 37.65 | 44.31 | 52.62 |
| 50 | 37.69               | 49.34 | 58.16 | 67.51 | 76.15 | 86.60 |

$\chi^2$  values

Fails to reject the null hypothesis  
 Rejects the null hypothesis

$df = n - 1$   
 $p$  = probability that the deviations from what was expected are due to pure chance